Problem 2

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Problem 1 *Let $f(x)$ and $g(x)$ be functions for which we only know the following:*

$$\int_1^4 f(x)\,dx = 7 \qquad \int_2^4 f(x)\,dx = 5 \qquad \int_1^4 g(x)\,dx = 2$$

Compute the following integrals, if possible. If it is not possible, give examples explaining why not.

(a) $\displaystyle\int_1^4 (8f(x) - 7g(x))\,dx$

Explanation.

$$\begin{aligned}\int_1^4 (8f(x) - 7g(x))\,dx &= 8\int_1^4 f(x)\,dx - 7\int_1^4 g(x)\,dx \\ &= 8(7) - 7(2) \\ &= 56 - 14 = 42\end{aligned}$$

(b) $\displaystyle\int_1^2 (-f(x))\,dx$

Explanation. First notice that

$$\int_1^4 f(x)\,dx = \int_1^2 f(x)\,dx + \int_2^4 f(x)\,dx$$

Therefore,

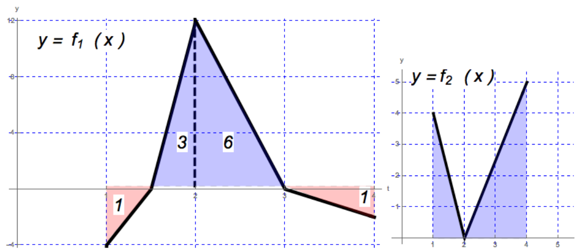
$$\int_1^4 f(x)\,dx - \int_2^4 f(x)\,dx = \int_1^2 f(x)\,dx$$

So

$$\begin{aligned}\int_1^2 (-f(x))\,dx &= -\int_1^2 f(x)\,dx \\ &= -\left(\int_1^4 f(x)\,dx - \int_2^4 f(x)\,dx\right) \\ &= -(7-5) = -2.\end{aligned}$$

(c) $\displaystyle\int_1^4 |f(x)|\,dx$

Explanation. We are not given enough information to solve this integral because we do not know the regions where f is positive or negative. Consider the following two functions $f_1(x)$ and $f_2(x)$:



Just using geometry, one can check that

$$\int_1^4 f_1(x)\,dx = 7 \qquad \int_2^4 f_1(x)\,dx = 5 \qquad \int_1^4 f_2(x)\,dx = 7 \qquad \int_2^4 f_2(x)\,dx = 5$$

and so both f_1 and f_2 satisfy the assumptions of f. But notice that

$$\int_1^4 |f_1(x)|\,dx = 11 \qquad \text{and} \qquad \int_1^4 |f_2(x)|\,dx = 7$$

These two examples demonstrate that we were not given enough information to solve this problem.

(d) $\displaystyle\int_1^4 (2 - x + f(x))\,dx$

Explanation. First notice that since the integral is linear over addition:

$$\int_1^4 (2 - x + f(x))\,dx = \int_1^4 2\,dx - \int_1^4 x\,dx + \int_1^4 f(x)\,dx = \int_1^4 2\,dx - \int_1^4 x\,dx + 7. \tag{1}$$

By using geometry, we can see that

$$\int_1^4 2\,dx = 2(4-1) = 6$$

$$\int_1^4 x\,dx = 1(4-1) + \frac{1}{2}(4-1)(4-1) = 3 + \frac{9}{2} = \frac{15}{2}.$$

Then substituting into equation (1) gives:

$$\int_1^4 (2 - x + f(x))\,dx = 6 - \frac{15}{2} + 7 = \frac{11}{2}.$$